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Studierichting: Informatica
Oefeningenreeks 4A nr. 44.7

$$\int \frac{x^2-43}{x^3+x+10}dx$$

Noemer ontbinden, $x=-2$:

$(-2)^3+(-2)+10=-8-2+10=0$

Dus $(x+2)$ is een factor:

Horner deling met $x=-2$:

$$\begin{array}{c|cccc} & 1 & 0 & 1 & 10 \\ -2 & & -2 & 4 & -10 \\ \hline & 1 & -2 & 5 & 0 \end{array}$$

Dus $x^3+x+10=(x+2)(x^2-2x+5)$

$x^3+x+10=(x+2)(x^2-2x+5)$

Partieelbreuken:

$$\frac{x^2-43}{(x+2)(x^2-2x+5)}=\frac{A}{x+2}+\frac{Bx+C}{x^2-2x+5}$$

$x^2-43=A(x^2-2x+5)+(Bx+C)(x+2)$

$$A=\left.\frac{x^2-43}{x^2-2x+5}\right|_{x=-2}=\frac{4-43}{4+4+5}=\frac{-39}{13}=-3$$

$$B(1+2i)+C \Rightarrow \left.\frac{x^2-43}{x+2}\right|_{x=1+2i}=\frac{(1+2i)^2-43}{(1+2i)+2}=\frac{1+4i-4-43}{3+2i}=\frac{-46+4i}{3+2i}$$

$$=\frac{(-46+4i)(3-2i)}{(3+2i)(3-2i)}=\frac{-138+92i+12i+8}{13}=\frac{-130+104i}{13}=-10+8i$$

$B(1+2i)+C=-10+8i$

$\Rightarrow 2B=8$

$\Rightarrow B=4$

$B+C=-10$

$\Rightarrow C=-14$

Dus:

$$\frac{x^2-43}{(x+2)(x^2-2x+5)} = \frac{-3}{x+2} + \frac{4x-14}{x^2-2x+5}$$

Integreren:

Eerste integraal:

$$\int \frac{-3}{x+2}dx = -3\ln|x+2|$$

Tweede integraal:

$$\int \frac{4x-14}{x^2-2x+5}dx = \int \frac{4x-4}{x^2-2x+5}dx - \int \frac{10}{(x-1)^2+4}dx$$

Eerste deel: $u = x^2-2x+5 \Rightarrow du = (2x-2)dx$:

$$\int \frac{4x-4}{x^2-2x+5}dx = 2\int \frac{du}{u} = 2\ln|x^2-2x+5|$$

Tweede deel: $t = x-1$:

$$\int \frac{10}{(x-1)^2+4}dx = \frac{10}{2}\operatorname{Bgtan}\left(\frac{x-1}{2}\right) = 5\operatorname{Bgtan}\left(\frac{x-1}{2}\right)$$

Alles samen:

$$\int \frac{x^2-43}{x^3+x+10}dx = -3\ln|x+2| + 2\ln(x^2-2x+5) - 5\operatorname{Bgtan}\left(\frac{x-1}{2}\right) + c$$

References